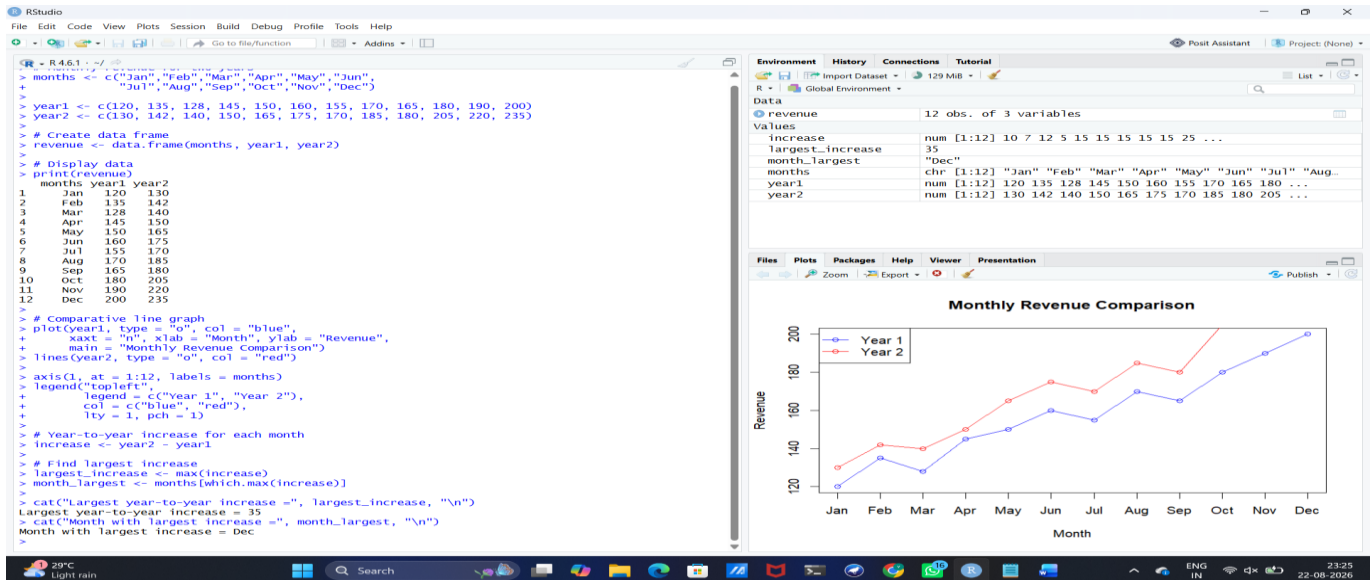
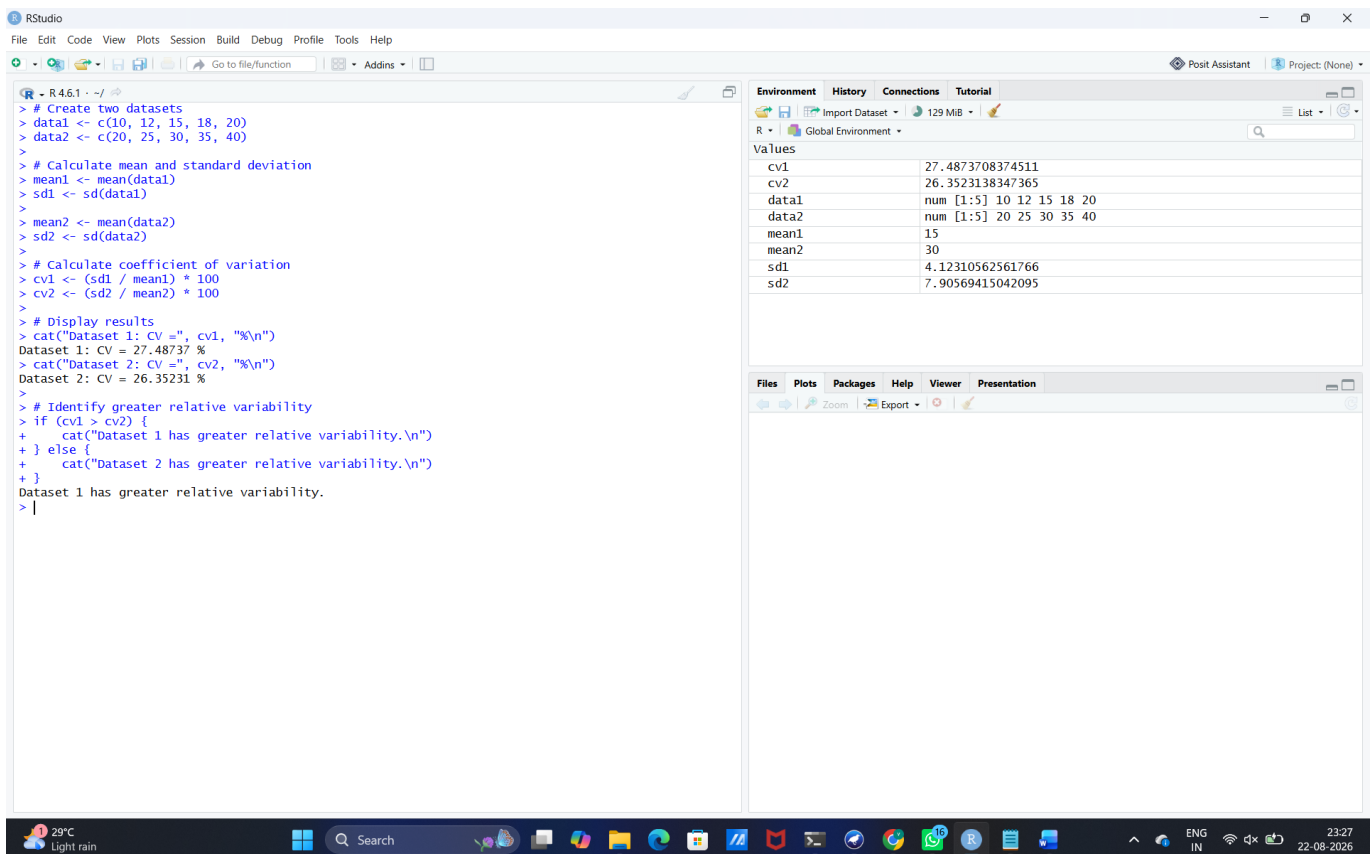


SET 18

1. R: Create monthly revenue for two years; draw a comparative line graph and find the largest year-to-year increase.



2. R: Calculate coefficient of variation for two datasets and identify greater relative variability.



3. WEKA: Create medical-screening ARFF data (BloodPressure, Cholesterol, Age, Risk); apply J48 and visualize.

Dataset: Medical Screening

```
@relation medical_screening
@attribute BloodPressure numeric
@attribute Cholesterol numeric
@attribute Age numeric
@attribute Risk {Low,High}
@data
118,170,25,Low
135,220,45,High
125,190,35,Low
145,240,55,High
130,210,42,High
115,165,28,Low
150,250,60,High
122,180,32,Low
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier chosen is 'J48 -C 0.25 -M 2'. The test options are set to 'Cross-validation' with 8 folds. The classifier output is displayed, showing the following results:

Number of Leaves : 2
Size of the tree : 3
Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	7	87.5 %
Incorrectly Classified Instances	1	12.5 %
Kappa statistic	0.75	
Mean absolute error	0.125	
Root mean squared error	0.3536	
Relative absolute error	22.5	%
Root relative squared error	63.6396	%
Total Number of Instances	8	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Low	0.750	0.000	1.000	0.750	0.857	0.775	0.875	0.875	Low
High	1.000	0.250	0.800	1.000	0.889	0.775	0.875	0.800	High
Weighted Avg.	0.875	0.125	0.900	0.875	0.873	0.775	0.875	0.838	

=== Confusion Matrix ===

```
a b <-- classified as
3 1 | a = Low
0 4 | b = High
```

The result list on the left shows '23:30:00 - trees.J48' as the selected model.

4. WEKA: Apply Logistic classification to the medical data and compare accuracy and ROC area with J48.

Dataset: Medical Screening – Logistic

Use the same Medical Screening ARFF dataset from Set 18 Q3.

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

Logistic -R 1.0E-8 -M -1 -num-decimal-places 4

Test options

Use training set

Supplied test set

Set...

Cross-validation

Folds

8

Percentage split

%

66

More options...

(Nom) Risk

Start

Stop

Result list (right-click for options)

23:30:00 - trees.J48

23:30:52 - functions.Logistic

Classifier output

=====

BloodPressure0.2508

Cholesterol0.4133

Age0.1443

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances8100%

Incorrectly Classified Instances00%

Kappa statistic1

Mean absolute error0.0616

Root mean squared error0.1724

Relative absolute error11.0912%

Root relative squared error31.0251%

Total Number of Instances8

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Low
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	High
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b <-- classified as

4 0 | a = Low

0 4 | b = High

Status

OK

Log

 x 0